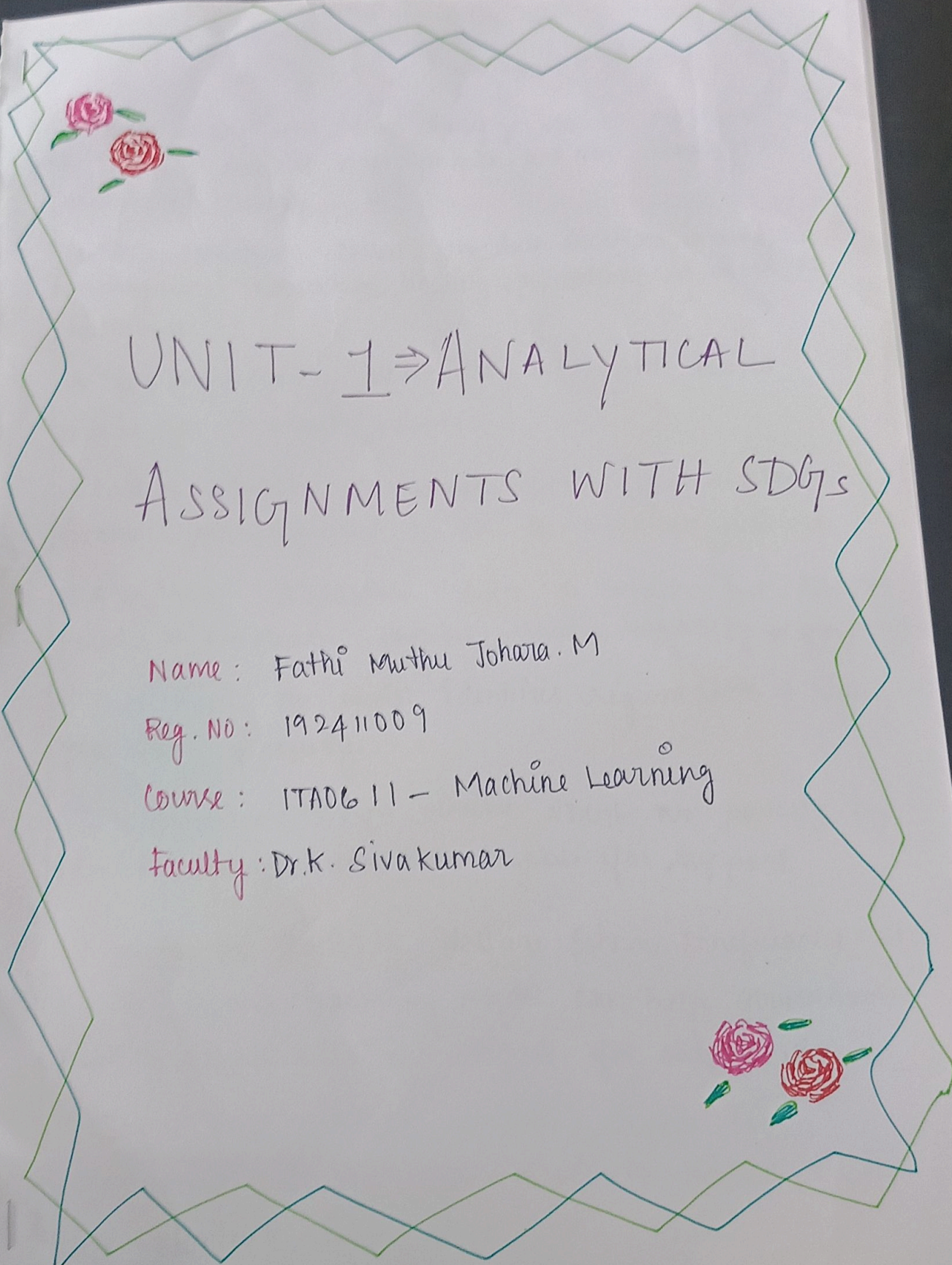




UNIT- 1 $\Rightarrow$ ANALYTICAL

ASSIGNMENTS WITH SDGs

Name : Fathi Muthu Johara. M

Reg. NO : 192411009

Course : ITAD611 - Machine Learning

Faculty : Dr. K. Sivakumar



ML to Identify Poverty-Prone Regions - SDG 1

Machine Learning can analyze socio-economic data to identify regions where people are more likely to experience poverty.

* **Data Collection**: Collect the data such as income, employment, education, housing, population and access to healthcare.

* **Data preprocessing**: clean the data and handle missing or incorrect values.

* **Feature selection**: Important factors such as low income, unemployment & poor education are selected.

* **ML Model**: Algorithms such as Decision Tree, Random Forest or k-Means can be used to classify or group regions.

* **Prediction**: The model identifies regions with a high probability of poverty.

SDG Connection: Helps achieve SDG 1- No poverty by identifying people who are eligible for support.

ML Model to Identify Welfare Scheme Beneficiaries - SDG 1

A Machine Learning model can help governments identify people who are eligible for welfare schemes.

* **Input Data**: collect income, family size, employment status, age, disability status and other eligibility information.

* **Preprocessing** : clean and organize the collected

* **Training** : use historical records of eligible and non-eligible beneficiaries to train the model.

* **Algorithm** : A Decision Tree or Random Forest can classify applicants as eligible

* **Prediction** : When a new application is given, the model predicts whether the person qualifies

SDG Connection : Supports SDG 1 - No poverty by improving access to government assistance.

3. ML for Crop Yield prediction - SDG 2

Machine learning can predict crop yield by analyzing historical agricultural data.

* **Data collection** : collect previous data on rainfall, temperature, soil type, fertilizer usage, irrigation and crop yield.

* **Feature selection** : Important agricultural factors are selected for prediction.

* **ML Model** : Algorithms such as Linear Regression, Decision Tree or Random Forest can be trained.

★ **Prediction**: The model predicts the expected crop yield for a particular season or region.

★ **Use**: Farmers can plan irrigation, fertilizer usage & crop selection based on prediction.

SDG Connection: Supports SDG 2 - Zero Hunger by Improving Agriculture.

ML for Crop Disease Detection using Leaf Images-SDG 2

Machine learning can detect crop diseases by analyzing photographs of plant leaves.

★ **Image Collection**: Collect images of healthy & diseased leaves.

★ **Preprocessing**: Resize and improve the images so that they can be processed by the model.

★ **Feature extraction**: The model learns features such as spots, discoloration and leaf patterns.

★ **ML Model**: A Convolutional Neural Network (CNN) can be trained using the leaf images.

★ **Action**: Farmers can detect disease early and take appropriate treatment measures.

SDG Connection: Supports SDG 2 - Zero Hunger by reducing crop losses and improving food production.

5. III To Reduce Food Wastage in Supermarkets - SDG 2

Machine learning can help super markets predict product demand & reduce excess food.

* **Data collection**: Use historical sales, seasonal trends, holidays, weather and product expiry data.

* **Demand prediction**: ML predicts how much of each food products will be sold.

* **Inventory management**: Stores can maintain stock according to predicted demand.

* **Expiry prediction**: The system can identify products that are approaching their expiry dates.

* **Result**: Less food is thrown away and supermarket losses are reduced.

SDG Connection: Supports SDG 2 - zero hunger by reducing food wastage and improving food utilization.